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# =====
# PRACTICAL 3
# Classification using Deep Feed Forward Network
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import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report,
    ConfusionMatrixDisplay
)

from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense, Input

# -----
# 1. Load Dataset
# -----
data = load_breast_cancer()

X = data.data
y = data.target

print("Dataset Shape:", X.shape)
print("Number of Classes:", len(np.unique(y)))

# -----
# 2. Split Dataset
# -----
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

# -----
# 3. Normalize Features
# -----
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# -----
# 4. Create Deep Neural Network
# -----
model = Sequential([
    Input(shape=(X_train.shape[1],)),

    Dense(64, activation='relu'),
    Dense(32, activation='relu'),
    Dense(16, activation='relu'),

    Dense(1, activation='sigmoid')
])

# -----
# 5. Compile
# -----
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

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model.summary()

# -----
# 6. Train
# -----
history = model.fit(
    X_train,
    y_train,
    validation_split=0.2,
    epochs=30,
    batch_size=32,
    verbose=1
)

# -----
# 7. Prediction
# -----
probabilities = model.predict(
    X_test,
    verbose=0
).flatten()

y_pred = (probabilities >= 0.5).astype(int)

# -----
# 8. Accuracy
# -----
accuracy = accuracy_score(
    y_test,
    y_pred
)

print("\nTest Accuracy:", round(accuracy, 4))

# -----
# 9. Classification Report
# -----
print("\nClassification Report:")
print(
    classification_report(
        y_test,
        y_pred,
        target_names=data.target_names
    )
)

# -----
# 10. Confusion Matrix
# -----
cm = confusion_matrix(
    y_test,
    y_pred
)

print("\nConfusion Matrix:")
print(cm)

ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=data.target_names
).plot()

plt.title("Confusion Matrix")
plt.show()

# -----
# 11. Accuracy Graph
# -----
plt.figure(figsize=(8, 5))

plt.plot(
    history.history['accuracy'],
    label='Training Accuracy'
)

plt.plot(

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history.history['val_accuracy'],  
label='Validation Accuracy'  
)  
  
plt.xlabel("Epoch")  
plt.ylabel("Accuracy")  
  
plt.title("Training and Validation Accuracy")  
  
plt.legend()  
plt.show()
```

